

Aryan Singh

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TECHNICAL SKILLS

Financial Analysis: Excel (PivotTables, INDEX/MATCH, Array Formulas), Financial Modeling, Variance Analysis, Risk Analysis

Data & Analytics: SQL (MySQL), Power BI (DAX), pandas, NumPy, scikit-learn, Statistical Analysis, Data Reconciliation

Programming: Python, C/C++, JavaScript, TypeScript, React, Next.js, FastAPI

Tools & Platforms: Git, GitHub, Jupyter Notebook, Streamlit, Vercel, PowerPoint

EXPERIENCE

Internal Auditor

Canadian National Exhibition

Aug. – Sep. 2026

Toronto, ON

- Audited **20+ concession operations** over an 18-day exhibition, verifying vendor-reported sales against on-site observation.
- Reconciled seller statements against audit records in Excel, escalating **10+ reporting exceptions** to the Internal Audit lead.
- Detected **25+ cash discrepancies** at gates and rides, classifying concession operators across four audit risk tiers by exposure.

Data Analyst

Focus on Nature

May – Aug. 2026

Guelph, ON

- Built a multi-tab Excel dashboard scoring and ranking **2,891 Ontario schools** and 103 organizations by socioeconomic need.
- Modeled a weighted socioeconomic score comparing school-level low-income and parental education rates against provincial levels of **9.25%** and 4.61%, using dual-key INDEX/MATCH array formulas and PivotTables.
- Verified **135 school records** against public sources manually, correcting errors that would have misdirected charity outreach.
- Presented findings for **9 Ontario regions** to the Board of Directors, prioritizing outreach for program planning.

PROJECTS

Robo-Advising Portfolio Optimizer | *Python, pandas, NumPy, yfinance*

[GitHub](#)

- Screened equities on currency, liquidity, and market capitalization, resolving cross-listed duplicates by trading volume.
- Filtered the **top 25% most volatile** names per sector, ranking the rest by correlation to an S&P 500 and TSX benchmark.
- Constructed a **\$1,000,000 CAD portfolio** matching sector weights under 40% and 15% position caps, net of FX and fees.

Market Stress & Crisis Simulator | *Next.js, TypeScript, React, Recharts*

[GitHub](#)

- Engineered a **stress-testing framework** replaying the 2008 financial crisis, COVID-19, and 2022 interest rate shock scenarios.
- Simulated portfolio performance across **US and TSX-listed equities** with daily, weekly, or monthly rebalancing.
- Computed **6 risk metrics** including maximum drawdown, Sharpe ratio, volatility, and time-to-recovery over 12–36 months.

EPS Drift Scanner | *Python, FastAPI, React, pandas, scikit-learn*

[GitHub](#)

- Developed a real-time scanner comparing **consensus Wall Street EPS estimates** against reported actuals across 10+ equities.
- Designed a **scikit-learn anomaly model** flagging earnings surprises that deviate from a company's historical pattern.
- Deployed a full-stack FastAPI and React dashboard surfacing **11 quarters** of per-equity earnings surprise data and alerts.

Ontario Education Gap Analysis | *Power BI, DAX*

[Video](#)

- Benchmarked Ontario's **72.05% primary enrolment rate** against the 94.75% Canadian national average in Power BI.
- Compared **6 Ontario regions** against 8 global regions, exposing a 46-point spread between the strongest and weakest.
- Automated regional drill-down using **DAX measures** and bookmark navigation within a single interactive report view.

LEADERSHIP

International Service Project

Shelter Them Poverty Relief

Mar. 2024

Kigali, Rwanda

- Raised **\$2,000+** through student-led fundraising and managed budget allocation across the full project timeline.
- Collaborated with a **15+ member team** to plan and deliver a community infrastructure project on time and under budget.

EDUCATION

University of Waterloo

Honours Bachelor of Computing and Financial Management (Co-op)

Waterloo, ON

- **Relevant Coursework:** Global Capital Markets, Financial Markets & Analytics, Financial Reporting, Probability & Statistics, Data Structures & Algorithms, Microeconomics